

FADEL BATAL

fadelbt@sas.upenn.edu | Philadelphia, PA | (267) 902-8790 | [LinkedIn](#) | [GitHub](#)

Research Interests: Multimodal foundation models in neuro-oncology; deep learning for medical imaging; uncertainty quantification; clinical AI translation.

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science | Philadelphia, PA 05/2025
Master of Science in Engineering, *Bioengineering*; Concentration in *Computational Medicine* GPA: 3.97
• **Master's Thesis:** *Non-Invasive and Segmentation-Free Prediction of Molecular Subtypes and Immune Clusters in Pediatric Low-Grade Gliomas via Transfer Learning* (Advised by Dr. Anahita Fathi Kazerooni).
Bachelor of Science in Engineering, *Bioengineering*; Minor in *Chemistry* GPA: 3.83
• **Coursework:** Scientific Computing, Introductory Statistics, Big Data Analytics, Applied Machine Learning, Deep Learning, Biological Data Science, Math for Engineering Applications, Advanced Biomedical Imaging Applications, Signal and Systems

PUBLICATIONS

† Denotes equal contribution.

Batal, F., et al. "Uncertainty-Aware Risk Stratification in Pediatric Low-Grade Glioma Using Multimodal Data." *American Journal of Neuroradiology*, published July 2026. doi:10.3174/ajnr.A9550.
Kazerooni, A.F.†, **Batal, F.**†, et al. "PedDx: An Uncertainty-Aware Deep Learning Framework for Pediatric Posterior Fossa Tumor Diagnosis With Agentic AI-Enabled Independent External Evaluation" *In preparation*, 2026.
Kazerooni, A.F.†, **Batal, F.**†, et al. "Uncertainty-Aware AI Assistance for Pediatric Posterior Fossa Tumor Diagnosis: A Multireader Study." *In preparation*, 2026.
Vunnavu, B., Lu, M., **Batal, F.**, Gandhi, D., Familiar, A., Chrysochoou, D., Storm, P.B., Resnick, A., Vossough, A., Nabavizadeh, A., Fathi Kazerooni, A. "Atlas-Based MRI Spatial Phenotyping in Pediatric Low-Grade Glioma" *In preparation*, 2026.
An, K., **Batal, F.**, Svyatskaya, S., and Nguyen, V. "Gene Therapy as a Promising Approach for the Underlying Causes of Autism Spectrum Disorder." *Berkeley Pharma Tech Journal of Medicine* 3, no. 1 (2023): 10–25.

RESEARCH EXPERIENCE

Center for Data-Driven Discovery in Biomedicine (D3B) – Children's Hospital of Philadelphia | Philadelphia, PA 07/2025 – Present
Imaging Research Data Analyst
• Designed and led, as **first author**, an uncertainty-aware deep learning model for risk stratification in pediatric low-grade glioma, integrating transfer-learned deep learning features with molecular and clinical data across **360 imaging and 493 molecular subjects** from the Children's Brain Tumor Network, achieving **Harrell's C-indices of 0.73 and 0.70** in train and test cohorts while reducing per-patient prediction-uncertainty by **~75%** (published, *American Journal of Neuroradiology*).
• Co-led, as **co-first author**, PedDx, a multi-institutional, segmentation-free deep-learning model for pre-operative differential diagnosis of pediatric posterior fossa tumors across **1,114 patients**, achieving up to **0.95 AUC-ROC** and **87%** external-validation **accuracy** with out-of-distribution analysis; evaluated in a multi-reader study with physicians across the U.S. to assess real-world diagnostic utility.
• Developed a clinical web application and **agentic-AI decision-support layer** for end-to-end posterior fossa tumor prediction; independently tested by a collaborating team in Australia, achieving **81% accuracy** across all histologies.
• Administer the unit's **Flywheel imaging database** and lead all pediatric brain-tumor imaging operations, including data ingestion, session management, preprocessing, tumor segmentation, quality control, and volumetric measurement, maintaining standardized, analysis-ready pipelines across 1,000+ cases that support research lab-wide.
• Mentor incoming master's students on their research projects, co-lead lab meetings with the PI, and serve as the lab's technical point of contact.
Data Science Intern 04/2024 – 05/2025
• Developed a segmentation-free, non-invasive deep-learning framework for pediatric brain-tumor molecular classification directly from MRI, achieving **91% AUC** and forming the basis of the RSNA 2025 oral presentation and master's thesis, with potential to reduce surgical tissue sampling and support personalized therapy selection.
Shorter Lab, Perelman School of Medicine | Undergraduate Researcher | Philadelphia, PA 08/2022 – 04/2024
• Analyzed TDP-43 protein misfolding under oxidative stress, quantifying aggregation kinetics (AUC, half-time) via dose-response curve fitting across GSSG concentrations, and demonstrating a **1.8-fold** increase in aggregation and a decline in Clip34 RNA therapeutic efficacy from 100% to 1–5% under stress.
Berkeley Pharma Tech | Medical Researcher Intern | Remote 08/2022 – 12/2022
• Co-authored a review article in *Berkeley Pharma Tech Journal of Medicine* analyzing the genetic basis of autism spectrum disorder and potential gene therapy treatments (DOI: [10.52243/bptjm.v1i4.38](https://doi.org/10.52243/bptjm.v1i4.38)).

PRESENTATIONS

Batal, F., et al. “A Segmentation-Free Deep Learning Framework for Pediatric Posterior Fossa Differential Diagnosis.” Poster, Society for Neuro-Oncology (SNO) Annual Meeting, Philadelphia, PA, Nov 2026.

Batal, F., et al. “Uncertainty-Aware Multimodal Survival Modeling in Pediatric Low-Grade Glioma.” Poster, Society for Neuro-Oncology (SNO) Annual Meeting, Philadelphia, PA, Nov 2026.

Batal, F. “Deep Learning for Differential Diagnosis of Pediatric Posterior Fossa Tumors.” Invited talk, Penn Brain Tumor Center, Philadelphia, PA, Aug 2026.

Batal, F., et al. “Non-Invasive and End-to-End Prediction of Molecular and Immune Subgroups of Pediatric Low-Grade Gliomas via Transfer Learning.” *Oral presentation*, Radiological Society of North America (RSNA) Annual Meeting, Chicago, IL, Nov 2025.

Batal, F., et al. “PADSSY: Peripheral Artery Disease Screening System.” *Poster*, Northeast Bioengineering Conference (NEBEC), Apr 2025.

Batal, F., et al. “PADSSY: Peripheral Artery Disease Screening System.” *Poster*, Senior Design Showcase, Design of Medical Devices (DMD) Conference, Minneapolis, MN, Apr 2025.

HONORS & AWARDS

- Summa Cum Laude, University of Pennsylvania (2025)
- Penn World Scholar (2021–2025)
- Emerging Medical Innovation Valuation, Design of Medical Devices Conference, Finalist 2025
- Student Design Showcase, Design of Medical Devices Conference, Finalist (7th place), 2025
- Dean’s List (2021–2022)
- Syrian National Chemistry Olympiad: Bronze Medalist (2018)

LEADERSHIP & VOLUNTEERING

University of Pennsylvania, Department of ESE | Teaching Assistant – Deep Learning 08/2025 – 12/2025

- Supported undergraduate deep learning coursework; graded assignments and provided technical feedback on final projects.

Weingarten Center | Organic Chemistry Tutor 01/2024 – 05/2024

- Tutored undergraduates in organic chemistry one-on-one, clarifying course concepts and designing custom worksheets and practice problems to prepare students for exams.

American Red Cross | Volunteer 05/2023 – 03/2024

- Led “Prepare with Pedro,” a K–2 fire- and hazard-safety education program in West Philadelphia schools, and delivered Hands-Only CPR training to employees across local companies.

Syrian Youth Empowerment | Mentor 05/2022 – 05/2025

- Supported Syrian high school students in college applications, authored research on Syrian education systems for U.S. admissions offices, and compiled scholarship resources, resulting in mentee admissions to top universities.

Science Olympiad at the University of Pennsylvania (SOUP) | Supervisor 09/2021 – 01/2023

- Designed and evaluated chemistry lab exams for Philadelphia-area high school teams competing in SOUP.

SELECTED PROJECTS

PADSSY: Peripheral Artery Disease Screening System | Python, Scikit-learn, Pandas, tkinter 08/2024 – 05/2025

- Built an ML-driven PAD screening solution using CDC cardiovascular datasets (400k+ entries), achieving **94%** diagnostic accuracy, and deployed the model in an EHR-inspired GUI for clinical usability.
- Co-engineered a portable, low-cost PAD diagnostic device involving dual blood-pressure circuit design, PCB prototyping, and signal-processing filters, achieving **97%** blood-pressure accuracy relative to FDA-approved devices.

Source and Syntax in News Headline Classification | Python, PyTorch, Transformers, Scikit-learn 03/2025 – 05/2025

- Developed a transformer- and LSTM-based NLP pipeline for news source classification (NBC vs. Fox); achieved **95% F1** with a POS-enhanced BiLSTM and conducted comparative EDA on linguistic and stylistic features to optimize model inputs.

Age-Adaptive Story Generation Model | PyTorch, Transformers, CUDA, Scikit-learn 11/2024 – 12/2024

- Fine-tuned a GPT-2 model to generate child-friendly stories from the TinyStories dataset, implementing a dual-task architecture with a classifier for age-appropriate content detection, achieving **99.92% test accuracy** after optimization.

SKILLS

Programming Languages: Python, SQL, MATLAB

Deep Learning & Computer Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV

Medical Imaging: nibabel, ITK, DICOM/NIfTI Pipelines, Image Registration, Tumor Segmentation, Flywheel

NLP & Agentic AI: Transformers, Agentic AI / LLM-Based Decision Support

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Software Development & Infrastructure: FastAPI, Streamlit, Linux, Git, Docker, slurm cluster